KU LEUVEN

MECANO
THE MECHANICS OF CANON FORMATION

Linguistic annotation using language models: Introduction

MECANO workshop Leuven

Workflows for Natural Language Processing

KU Leuven, November 17-19, 2025

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Summary

Linguistic annotation: introduction

- What is linguistic annotation?
- Annotation layers
- Annotation styles: what standard?
  - Universal Dependencies (UD)

Questions?

What kind of linguistic resources?

- Annotated corpora: manual vs automatic
- How to use annotated corpora

Questions?

How is annotation performed?

- Manual
- Rule-based
- Statistics/machine learning
- Deep Neural Netwoks

Questions?

How to annotate your own text?

- Hands-on session:
  - User interface models
  - Python notebook
  - Evaluation of the results

Questions?

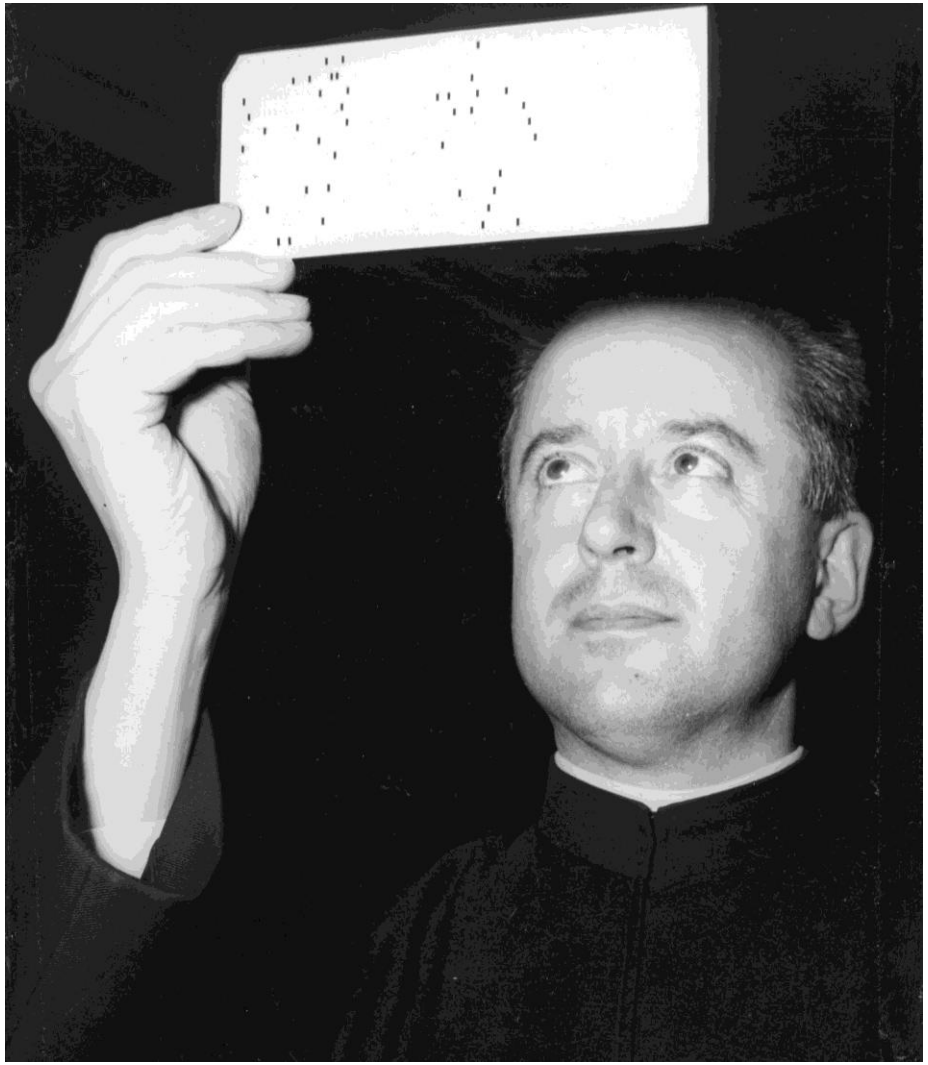
Linguistic annotation

Introduction

Where it all started…

Father Roberto Busa (1913-2011) and the *Index Thomisticus*:

from a lemmatized concordance to a syntactically annotated corpus, *the Index Thomisticus Treebank.*



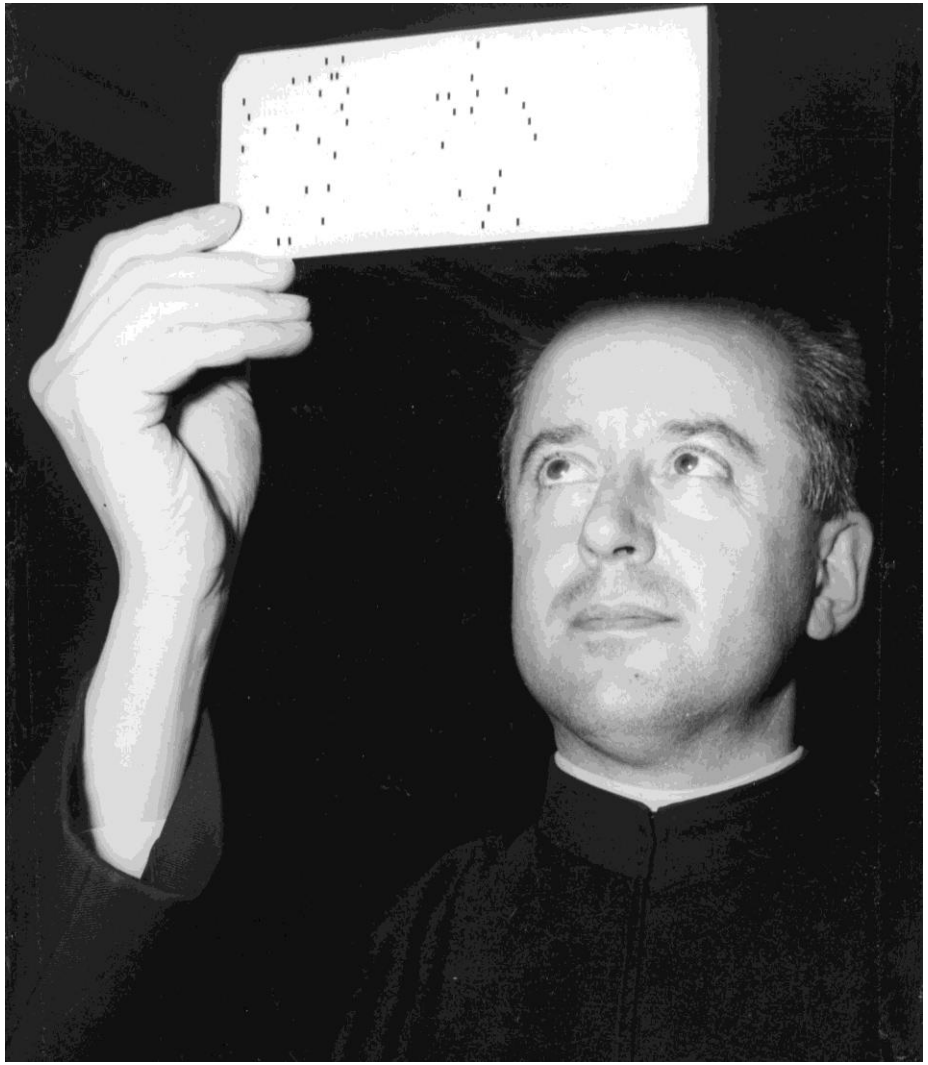
Father Busa holding a punched card – image available online

Where it all started…

Father Roberto Busa (1913-2011) and the *Index Thomisticus*:

from a lemmatized concordance to a syntactically annotated corpus, *the Index Thomisticus Treebank.*

Fun fact: he was my mother's teacher in Milan!



Father Busa holding a punched card – image available online

Linguistic annotation: what is all about?

- Corpus annotation: **enhancing raw text with metadata** of different kind.
- Different types of annotation (linguistic, semantic, multimodal…)
- Linguistic annotation: enhancing the text with linguistic information.
- **Layers of annotation**: from tokenization...

…to infinity and beyond!



These slides are adapted from materials by Margherita Fantoli, Marco Passarotti and Rachele Sprugnoli.

(Some possible) annotation layers

Tokenization

Phonetic and prosodic annotation (spoken corpora)

Lemmatization

Part-of-Speech tagging and morphological analysis

Syntactical dependencies → treebanks

Anaphora and ellypsis resolution

Semantic annotation (NER, Wordnet, Semantic Role Labeling…)

Multimodal annotation (gesture, facial expressions, perceptible context…)

Pipeline example

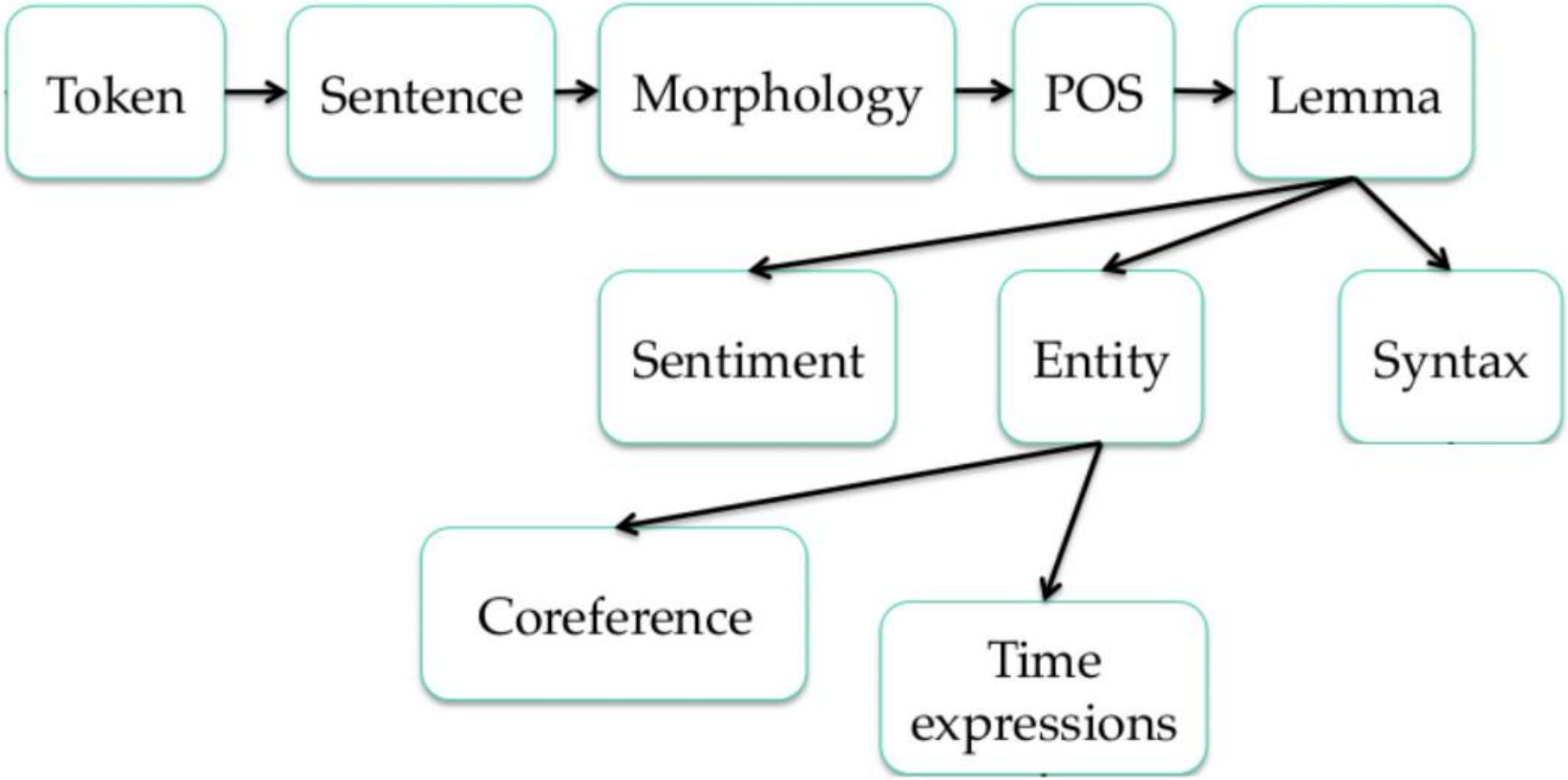


Image taken from Rachele Sprugnoli. (2022, July 5). Natural Language Processing Methods. Zenodo. https://doi.org/10.5281/zenodo.6798390

Linguistic layers of annotation: the essentials

Tokenization	Converting a sequence of characters (i.e. your text) into a sequence of lexical tokens
Lemmatization	Assigning each token to its citation form
Part-of-Speech tagging and morphological analysis	Assigning Part-of-Speech tags and analyzing morphological features of lexical tokens
Syntactical dependencies → treebanks	Organizing the text in syntactic graphs (= treebanks), representing dependency relations between the tokens

Linguistic layers of annotation: the essentials

Tokenization

Converting a sequence of characters (i.e. your text) into a sequence of lexical tokens

Tantaene animis caelestibus irae?

How many tokens?

technique		
white spaces	Tantaene | animis | caelestibus | irae?	4
removing punctuation	Tantaene | animis | caelestibus | irae	4
Separating punctuation	Tantaene | animis | caelestibus | irae | ?	5
Splitting clitics	Tantae | ne | animis | caelestibus | irae | ?	6

Linguistic layers of annotation: the essentials

Lemmatization — Assigning each token to its citation form

Tantaene animis caelestibus irae?

token	Lemma
tantae	tantus
ne	ne
animis	animus
caelestibus	caelestis
irae	ira

Linguistic layers of annotation: the essentials

Lemmatization

Assigning each token to its citation form

CAELESTIS: how many lemmas?

Lemma	PoS
caelestis	proper noun
caelestis	common noun
caelestis celestis coelestis	adjective

https://lila-erc.eu/#page-top

Linguistic layers of annotation: the essentials

Part-of-Speech tagging and morphological analysis

Assigning Part-of-Speech tags and analyzing morphological features of lexical tokens

Tantaene animis caelestibus irae?

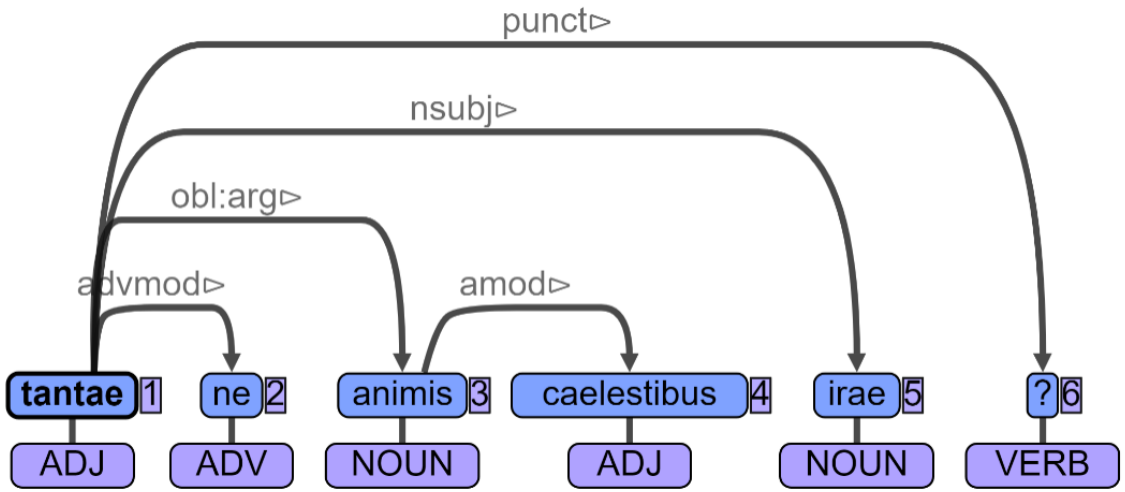
token	Part-of-Speech
tantae	adjective? determiner?
ne	particle? adverb?
animis	noun
caelestibus	adjective? proper noun? noun?
irae	noun

Linguistic layers of annotation: the essentials

Syntactical dependencies → treebanks

Organizing the text in syntactic graphs (= treebanks), representing dependency relations between the tokens

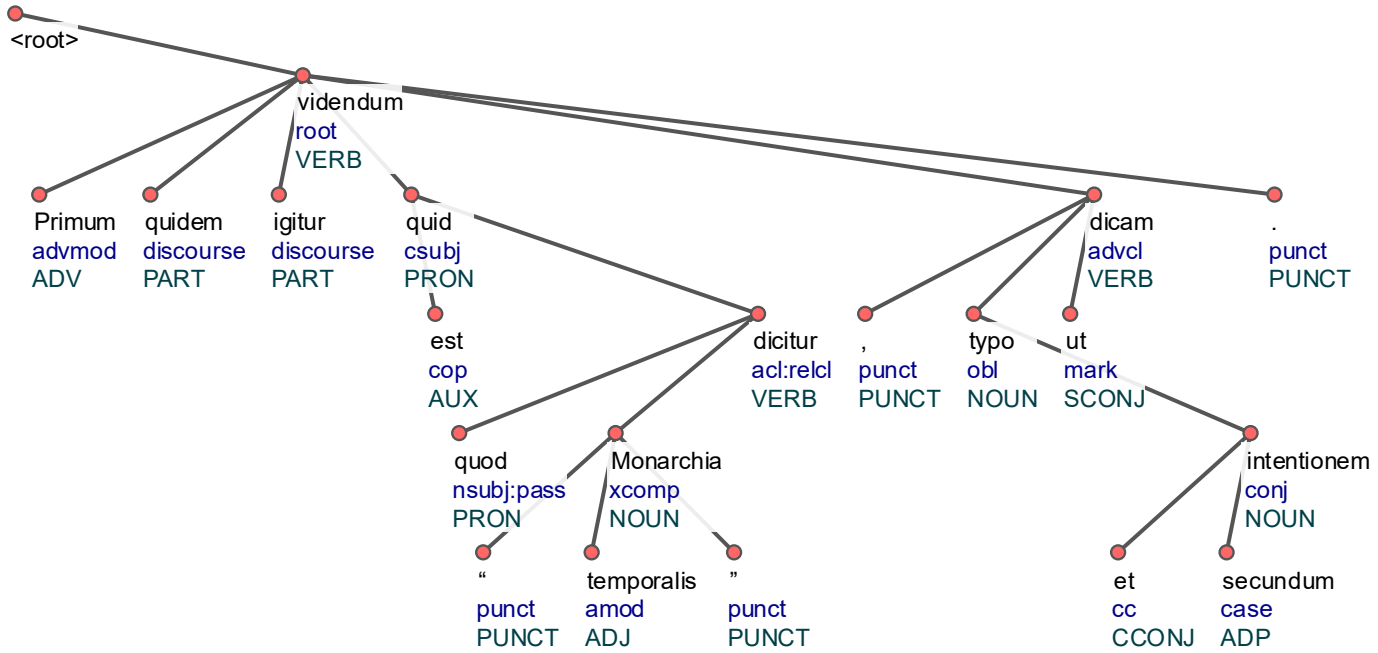
Tantaene animis caelestibus irae?



Linguistic layers of annotation: the essentials

Syntactical dependencies → treebanks

Organizing the text in syntactic graphs (= treebanks), representing dependency relations between the tokens



Annotation styles

Different projects follow different conventions for linguistic annotation

Despite many common points, they differ on some crucial choices, such as:

- How to annotate “particles” (enim, autem, uero etc.)?
- How to treat words used in a different function, e.g. nominalized adjectives etc.?
- How to deal with spelling variation and irregular declensions?

For classical studies we find, for instance:

- Use and combination of different dictionaries as lemmatization reference (Forcellini, L&S etc.)
- Different tagsets for Part-of-Speech and morphology
- Different parsing styles (constituent vs dependency)

Annotation styles

Different projects follow different conventions for linguistic annotation

Despite many co as:

- How to ann
- How to trea djectives etc.?
- How to dea

For classical stu

- Use and co erence (Forcellini,
- Different ta
- Different parsing styles (constituent vs dependency)

Imagine if we want to compare multilingual datasets!

A possible standard: Universal Dependencies (UD)

"Universal Dependencies (UD) is a project that is developing **cross-linguistically consistent** treebank annotation **for many languages**, with the goal of facilitating multilingual parser development, cross-lingual learning, and parsing research from a language typology perspective."

(https://universaldependencies.org/introduction.html)

- cross-linguistically consistent
- building on existing standards
- complementing, and not replacing, language-specific schemes
- open community effort

A possible standard: Universal Dependencies (UD)

Standard Part-of-Speech tags (**UPOS tags**): universaldependencies.org/u/Part-of-Speech/index.html

Open class words	Closed class words	Other
ADJ	ADP	PUNCT
ADV	AUX	SYM
INTJ	CCONJ	X
NOUN	DET	
PROPN	NUM	
VERB	PART	
	PRON	
	SCONJ	

A possible standard: Universal Dependencies (UD)

Standard morphological features: https://universaldependencies.org/u/feat/index.html.

Lexical features*	Inflectional features*	
	*Nominal**	*Verbal**
PronType	Gender	VerbForm
NumType	Animacy	Mood
Poss	NounClass	Tense
Reflex	Number	Aspect
Other	Case	Voice
Abbr	Definite	Evident
Typo	Deixis	Polarity
Foreign	DeixisRef	Person
ExtPos	Degree	Polite
		Clusivity

A possible standard: Universal Dependencies (UD)

Standard dependency relations: https://universaldependencies.org/u/dep/index.html

	Nominals	Clauses	Modifier words	Function Words
Core arguments	nsubj obj iobj	csubj ccomp xcomp		
Non-core dependents	obl vocative expl dislocated	advcl	advmod* discourse	aux cop mark
Nominal dependents	nmod appos nummod	acl	amod	det clf case
Coordination	**MWE**	**Loose**	**Special**	**Other**
conj cc	fixed flat compound	list parataxis	orphan goeswith reparandum	punct root dep

A possible standard: Universal Dependencies (UD)

Standard file format CoNLL-U: https://universaldependencies.org/format.html.

```
# sent_id = 1
# text = They buy and sell books.
1   They     they    PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and     CCONJ   CC     _                                   4   cc      4:cc              _
4   sell     sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book    NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .       PUNCT   .      _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I       I       PRON    PRP    Case=Nom|Number=Sing|Person=1     2   nsubj   _   _
2   have    have    VERB    VBP    Number=Sing|Person=1|Tense=Pres   0   root    _   _
3   no      no      DET     DT     PronType=Neg                      4   det     _   _
4   clue    clue    NOUN    NN     Number=Sing                       2   obj     _   SpaceAfter=No
5   .       .       PUNCT   .      _                                 2   punct   _   _
```

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1   They     they    PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and     CCONJ   CC     _                                   4   cc      4:cc              _
4   sell     sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book    NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .       PUNCT   .      _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I        I       PRON    PRP    Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have     have    VERB    VBP    Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no       no      DET     DT     PronType=Neg                        4   det     _   _
4   clue     clue    NOUN    NN     Number=Sing                         2   obj     _   SpaceAfter=No
5   .        .       PUNCT   .      _                                   2   punct   _   _
```

ID: word index

A possible standard: Universal Dependencies (UD)

Standard file format CoNLL-U: https://universaldependencies.org/format.html.

```
# sent_id = 1
# text = They buy and sell books.
1   They    they    PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy     buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and     and     CCONJ   CC     _                                   4   cc      4:cc              _
4   sell    sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books   book    NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .       .       PUNCT   .      _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I       I       PRON    PRP    Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have    have    VERB    VBP    Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no      no      DET     DT     PronType=Neg                        4   det     _   _
4   clue    clue    NOUN    NN     Number=Sing                         2   obj     _   SpaceAfter=No
5   .       .       PUNCT   .      _                                   2   punct   _   _
```

FORM: word form or symbol

A possible standard: Universal Dependencies (UD)

Standard file format CoNLL-U: https://universaldependencies.org/format.html.

```
# sent_id = 1
# text = They buy and sell books.
1   They     they    PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and     CCONJ   CC     _                                   4   cc      4:cc              _
4   sell     sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book    NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .       PUNCT   .      _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I        I       PRON    PRP    Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have     have    VERB    VBP    Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no       no      DET     DT     PronType=Neg                        4   det     _   _
4   clue     clue    NOUN    NN     Number=Sing                         2   obj     _   SpaceAfter=No
5   .        .       PUNCT   .      _                                   2   punct   _   _
```

LEMMA: lemma or stem of word form

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1   They     they    PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and     CCONJ   CC     _                                   4   cc      4:cc              _
4   sell     sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book    NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .       PUNCT   .      _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I        I       PRON    PRP    Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have     have    VERB    VBP    Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no       no      DET     DT     PronType=Neg                        4   det     _   _
4   clue     clue    NOUN    NN     Number=Sing                         2   obj     _   SpaceAfter=No
5   .        .       PUNCT   .      _                                   2   punct   _   _
```

UPOS: universal Part-of-Speech tag

A possible standard: Universal Dependencies (UD)

Standard file format CoNLL-U: https://universaldependencies.org/format.html.

```
# sent_id = 1
# text = They buy and sell books.
1   They     they     PRON    PRP   Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy      VERB    VBP   Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and      CCONJ   CC    _                                   4   cc      4:cc              _
4   sell     sell     VERB    VBP   Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book     NOUN    NNS   Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .        PUNCT   .     _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I        I        PRON    PRP   Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have     have     VERB    VBP   Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no       no       DET     DT    PronType=Neg                        4   det     _   _
4   clue     clue     NOUN    NN    Number=Sing                         2   obj     _   SpaceAfter=No
5   .        .        PUNCT   .     _                                   2   punct   _   _
```

XPOS: Optional language-specific (or treebank-specific) part-of-speech / morphological tag

A possible standard: Universal Dependencies (UD)

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```
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# text = They buy and sell books.
1   They     they    PRON    PRP    Case=Nom|Number=Plur                  2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres       0   root    0:root            _
3   and      and     CCONJ   CC     _                                     4   cc      4:cc              _
4   sell     sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres       2   conj    0:root|2:conj     _
5   books    book    NOUN    NNS    Number=Plur                           2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .       PUNCT   .      _                                     2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I        I       PRON    PRP    Case=Nom|Number=Sing|Person=1     2   nsubj   _   _
2   have     have    VERB    VBP    Number=Sing|Person=1|Tense=Pres   0   root    _   _
3   no       no      DET     DT     PronType=Neg                      4   det     _   _
4   clue     clue    NOUN    NN     Number=Sing                       2   obj     _   SpaceAfter=No
5   .        .       PUNCT   .      _                                 2   punct   _   _
```

FEATS: List of morphological features

A possible standard: Universal Dependencies (UD)

Standard file format CoNLL-U: https://universaldependencies.org/format.html.

```
# sent_id = 1
# text = They buy and sell books.
1   They     they    PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and     CCONJ   CC     _                                   4   cc      4:cc              _
4   sell     sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book    NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .       PUNCT   .      _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I        I       PRON    PRP    Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have     have    VERB    VBP    Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no       no      DET     DT     PronType=Neg                        4   det     _   _
4   clue     clue    NOUN    NN     Number=Sing                         2   obj     _   SpaceAfter=No
5   .        .       PUNCT   .      _                                   2   punct   _   _
```

HEAD: Head of the current word, which is either a value of ID or zero (0).

A possible standard: Universal Dependencies (UD)

Standard file format CoNLL-U: https://universaldependencies.org/format.html.

```
# sent_id = 1
# text = They buy and sell books.
1   They     they    PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and     CCONJ   CC     _                                   4   cc      4:cc              _
4   sell     sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book    NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .       PUNCT   .      _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I        I       PRON    PRP    Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have     have    VERB    VBP    Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no       no      DET     DT     PronType=Neg                        4   det     _   _
4   clue     clue    NOUN    NN     Number=Sing                         2   obj     _   SpaceAfter=No
5   .        .       PUNCT   .      _                                   2   punct   _   _
```

DEPREL: Universal dependency relation to the HEAD.

A possible standard: Universal Dependencies (UD)

Standard file format CoNLL-U: https://universaldependencies.org/format.html.

```
# sent_id = 1
# text = They buy and sell books.
1   They     they     PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy      VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and      CCONJ   CC     _                                   4   cc      4:cc              _
4   sell     sell     VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book     NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .        PUNCT   .      _                                   2   punct   2:punct           _

# sent_id = 2
# text = I have no clue.
1   I        I        PRON    PRP    Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have     have     VERB    VBP    Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no       no       DET     DT     PronType=Neg                        4   det     _   _
4   clue     clue     NOUN    NN     Number=Sing                         2   obj     _   SpaceAfter=No
5   .        .        PUNCT   .      _                                   2   punct   _   _
```

DEPS: Enhanced dependency graph in the form of a list of head-deprel pairs.

A possible standard: Universal Dependencies (UD)

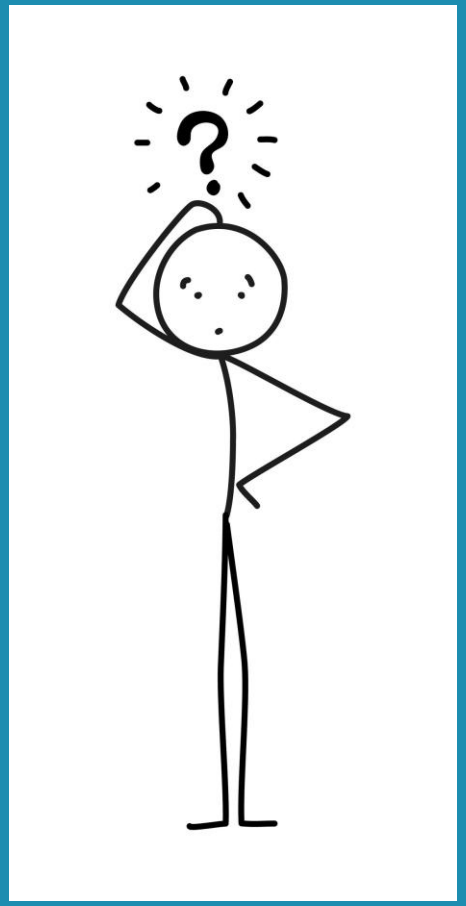
Standard file format CoNLL-U: https://universaldependencies.org/format.html.

```
# sent_id = 1
# text = They buy and sell books.
1   They     they    PRON    PRP    Case=Nom|Number=Plur                2   nsubj   2:nsubj|4:nsubj   _
2   buy      buy     VERB    VBP    Number=Plur|Person=3|Tense=Pres     0   root    0:root            _
3   and      and     CCONJ   CC     _                                   4   cc      4:cc              _
4   sell     sell    VERB    VBP    Number=Plur|Person=3|Tense=Pres     2   conj    0:root|2:conj     _
5   books    book    NOUN    NNS    Number=Plur                         2   obj     2:obj|4:obj       SpaceAfter=No
6   .        .       PUNCT   .      _                                   2   punct   2:punct           _

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# text = I have no clue.
1   I        I       PRON    PRP    Case=Nom|Number=Sing|Person=1       2   nsubj   _   _
2   have     have    VERB    VBP    Number=Sing|Person=1|Tense=Pres     0   root    _   _
3   no       no      DET     DT     PronType=Neg                        4   det     _   _
4   clue     clue    NOUN    NN     Number=Sing                         2   obj     _   SpaceAfter=No
5   .        .       PUNCT   .      _                                   2   punct   _   _
```

MISC: Any other annotation.

Any questions at this point?



Linguistic resources

Available annotated corpora and how to use them

What kind of linguistic resources?

Textual corpora

- Raw text
- Annotated text:

1. Manually
2. Automatically

Ontologies

- Formal and explicit representation of a concept
- Relations between concepts

Lexica and dictionaries

- Annotated:
  e. g. valency lexicons

What kind of linguistic resources?

Textual corpora

- Raw text
- Annotated text:

1. Manually
2. Automatically

Ontologies

- Formal and explicit representation of a concept
- Relations between concepts

Lexica and dictionaries

- Annotated:
  e. g. valency lexicons

Manually annotated corpora

LASLA corpus (Latin):

- 1.7 million syntactic words already published
- Classical literary texts
- Lemmatization & morphology
- Specific annotation schema (not UD!)
- https://www.lasla.uliege.be/cms/c_11821932/fr/lasla-lasla-dataverse

Manually annotated corpora

PROIEL (Pragmatic Resources of Old Indo-European Languages)

- Around 365 000 Latin syntactic words
- Around 250 000 Ancient Greek syntactic words
- Latin: Caesar, Cicero, Jerome's New Testament..
- Ancient Greek: Herodotus, Greek New Testament…
- Classical Armenian, Gothic and Old Church Slavonic
- Initially using Perseus formalism, now available as UD
- https://github.com/proiel/proiel-treebank

Manually annotated corpora

Ancient Greek and Latin Dependency Treebank (AGLDT, Perseus)

- Selection of Greek and Latin classical authors
- Initially using Perseus formalism → biggest part
- Now partially available as UD → small part
- https://github.com/PerseusDL/treebank_data/tree/master

Manually annotated corpora

Index Thomisticus Treebank (ITTB):

- 450 000 syntactic words
- Data are taken from *Index Thomisticus*, that contains all the works of Thomas Aquinas with lemmas, morphology and Part-of-Speech tagging
- Treebank: *Summa Contra Gentiles*
- Initially using Perseus formalism, now converted to UD
- https://itreebank.marginalia.it/
- https://github.com/UniversalDependencies/UD_Latin-ITTB/blob/master/README.md

Manually annotated corpora

LLCT (Late Latin Charter Treebank)

- 521 Early Medieval Latin charters written in Tuscany, 8th-9th centuries
- 242 411 tokens
- Initially using Perseus formalism, now available as UD
- https://github.com/UniversalDependencies/UD_Latin-LLCT

Manually annotated corpora

UDante:

- Latin works by Dante (*De vulgari eloquentia*, *Monarchia*, *Letters*, *Questio de aqua et terra*, *Eclogues*)
- Around 55 000 tokens
- Natively UD formalism
- https://github.com/UniversalDependencies/UD_Latin-UDante

Manually annotated corpora

CIRCSE treebank (Latin):

- Work in progress
- Currently contains Seneca *Hercules Furens*, Seneca *Agamemnon*, Tacitus *Germania*, Seneca *Oedipus*
- Around 23 000 syntactical words
- Texts received from LASLA with lemmatization, morphology and Part-of-Speech tags → converted to UD and manually revised
- Syntax following UD → from scratch
- https://github.com/UniversalDependencies/UD_Latin-CIRCSE/blob/master/README.md

Automatically annotated corpora

GLAUX Corpus (Ancient Greek):

- 20M tokens
- currently 8th century BC - roughly 4th century AD
- lemmas, morphology, syntactic dependencies and now animacy and word senses
- automatically annotated for morphology and syntax
- queryable interface
- https://glaux.be/
- https://github.com/alekkeersmaekers/glaux

Automatically annotated corpora

Corpus Latin antiquité et antiquité tardive lemmatisé:

- 21M tokens
- Classical and Late Latin text
- Annotated with the LASLA model
- Lemmatization, Part-of-Speech tags and morphological features
- https://github.com/lascivaroma/latin-lemmatized-texts/tree/0.1.2

Automatically annotated corpora

Opera Graeca Adnotata (OGA):

- 40M tokens
- Lemmatization, Part-of-Speech tags and morphological features
- Integration of CTS URNs
- IPA transcription
- Queryable interface
- https://varro.informatik.uni-leipzig.de/oga/en/

Automatically annotated corpora

Opera Latina Adnotata:

- 17M tokens
- Lemmatization, Part-of-Speech tags and morphological features
- Integration of CTS URNs
- IPA transcription
- https://zenodo.org/records/15183688

Other corpora (English and French)

UD English EWT:

- 250.000 tokens
- Lemmatization (automatic), Part-of-Speech tags, morphological features and syntax (manual)
- https://universaldependencies.org/treebanks/en_ewt/index.html

- More English UD annotated corpora: https://universaldependencies.org/

Other corpora (English and French)

UD French GSD:

- 400.000 tokens
- Lemmatization (automatic), Part-of-Speech tags, morphological features and syntax (semi-automatic).
- Originally annotated in non-UD style, later automatically converted.
- https://universaldependencies.org/treebanks/fr_gsd/

- More French UD annotated corpora: https://universaldependencies.org/

How to visualize and query the annotated corpora?

- **CoNLL-U viewer**: http://universaldependencies.org/conllu_viewer.html. Upload your file and visualize it.

- Some corpora have their **own interface**:
  - GLAUx: https://glaux.be/search.php
  - OGA: https://annis.varro.informatik.uni-leipzig.de/oga020

- Available infrastractures for querying UD annotated corpora:
  - **GrewMatch**: https://match.grew.fr/
  - **TüNDRA** (TiGer Search query language): https://weblicht.sfs.uni-tuebingen.de/Tundra/

TüNDRA -TiGer Search Query Language

- Very clear documentation: https://weblicht.sfs.uni-tuebingen.de/Tundra/help
- Basics:

  - Values: in **square brackets [ ]**
  - Example: [lemma="have"]

  - Variables:
    - **token**
    - **lemma**
    - **pos**
    - **edge** (= dependency relation)

TüNDRA -TiGer Search Query Language

- **Word order**-based relations:

1. [lemma = "habeo"] **.** [token = "res"] → **precedes immediately**
2. [lemma = "dico"] **.*** [lemma= "loquor"] → **precedes NOT immediately**

TüNDRA -TiGer Search Query Language

- **Syntax-**based relations:

1. [pos = "ADJ"] **>** [pos="NOUN"] → **heads directly**: noun depending on an adjective
2. [token = /.*as/] **>*** [pos="ADJ"] → **heads not directly**: token ending with –as heads not directly an adjective.
   Note: You can replace * with a number (>2 means heads at distance 2)
3. [pos="ADV"] **$** [pos="ADV"] → **siblings**: two adverbs on the same syntactic level
4. [lemma ="habeo"] **> obj** [pos = "DET"] → **labeled dependency**: a determiner depending with a object relation on the lemma habeo

TüNDRA -TiGer Search Query Language

- Supports **Regular Expressions** (between / /):

1. [token=/form.*/] → .* = zero or plus occurrences of any character.

- Supports **Boolean values**:
  - `!: negation (NOT)`
  - `&: conjunction (AND)`
  - `|: disjunctions (OR)`

1. [pos ="CCONJ"] | [pos="SCONJ"] → coordinating conjunction OR subordinating conjunction

- Supports use of **variables**:

1. #1: [pos="NOUN"] > #2: [pos="ADJ"] & #1 . [lemma="res"]

TüNDRA – A query example (1)

Let’s download a sample from the UDante corpus: **la_udante-ud-train.conllu.**

https://weblicht.sfs.uni-tuebingen.de/Tundra/

#1: [pos ="NOUN"]

Constraints:

- A noun (#1)

TüNDRA – A query example (1)

Let’s download a sample from the UDante corpus: **la_udante-ud-train.conllu.**

https://weblicht.sfs.uni-tuebingen.de/Tundra/

#1: [pos="NOUN"] > #2: [pos="ADJ"]

Constraints:

- A noun (#1) that governs (>) an adjective (#2)

TüNDRA – A query example (1)

Let's download a sample from the UDante corpus: **la_udante-ud-train.conllu.**

https://weblicht.sfs.uni-tuebingen.de/Tundra/

```
#1: [pos="NOUN"] > #2: [pos="ADJ"] & #1 . #2
```

Constraints:

- A noun (#1) that governs (>) an adjective (#2)
- The noun (#1) must precede immediately (.) the adjective (#2)

TüNDRA – A query example (1)

Let's download a sample from the UDante corpus: **la_udante-ud-train.conllu.**

https://weblicht.sfs.uni-tuebingen.de/Tundra/

```
#1: [pos="NOUN"] > #2: [pos="ADJ"] & #1 . #2 & #3:[lemma="dico"]
```

Constraints:

- A noun (#1) that governs (>) an adjective (#2)
- The noun (#1) must precede immediately (.) the adjective (#2)
- The lemma *dico* (#3)

TüNDRA – A query example (1)

Let's download a sample from the UDante corpus: **la_udante-ud-train.conllu**.

https://weblicht.sfs.uni-tuebingen.de/Tundra/

```
#1: [pos="NOUN"] > #2: [pos="ADJ"] & #1 . #2 & #3:[lemma="dico"] &
![lemma="loquor"]
```

Constraints:

- A noun (#1) that governs (>) an adjective (#2)
- The noun (#1) must precede immediately (.) the adjective (#2)
- The lemma *dico* (#3)
- NOT (!) the lemma *loquor*

TüNDRA – A query example (2)

Let’s download a sample from the UDante corpus: **la_udante-ud-train.conllu.**

https://weblicht.sfs.uni-tuebingen.de/Tundra/

#1: [pos = "NOUN" & lemma = /am.*/]

Constraints:

- A noun the lemma of which must start with am- (#1)

TüNDRA – A query example (2)

Let’s download a sample from the UDante corpus: **la_udante-ud-train.conllu.**

https://weblicht.sfs.uni-tuebingen.de/Tundra/

```
#1: [pos = "NOUN" & lemma = /am.*/] $ #2:[pos = "NOUN"]
```

Constraints:

- A noun the lemma of which must start with am- (#1)
- The noun (#1) must be sibling of another noun (#2)

TüNDRA – A query example (2)

Let's download a sample from the UDante corpus: **la_udante-ud-train.conllu.**

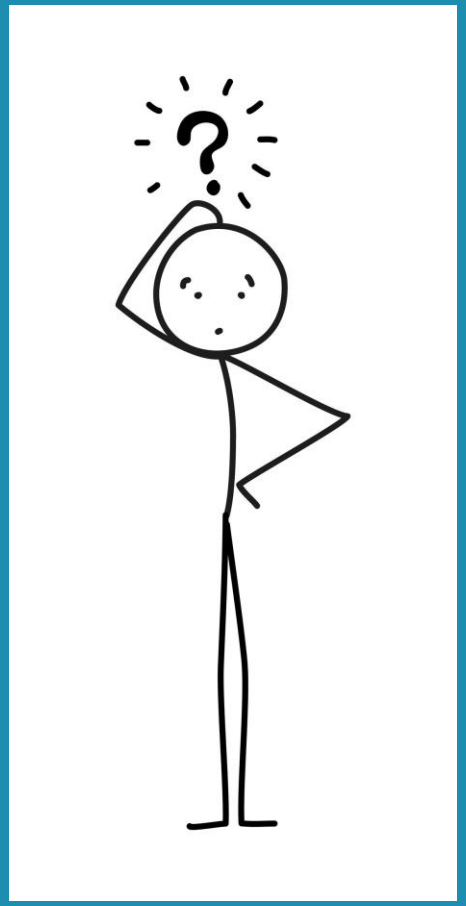
https://weblicht.sfs.uni-tuebingen.de/Tundra/

```
#1: [pos= "NOUN" & lemma = /am.*/] $ #2:[pos = "NOUN"] & #1 >nmod #3:
[token=/.*(is|ae|i|us|ei)/]
```

Constraints:

- A noun the lemma of which must start with am- (#1)
- The noun (#1) must be sibling of another noun (#2)
- The noun (#1) must govern with a relationship of nominal modifier a token (#3) that ends with –is, -ae, -i, -us or –ei.

Any questions at this point?



How is annotation performed?

Manually vs Automatically

Manual annotation: IAA agreement

- Manual annotation: GOLD data → **accurate** and **complete** annotation
- However, not as easy as it may sound

Inter-Annotator Agreement (IAA)

Agreement between at least two annotators on the same text

Broad agreement between the annotators means:

1. Valid annotation scheme
2. High quality of annotated data

Manual annotation: the problem

- Manual annotation: GOLD data → **accurate** and **complete** annotation
- Howe

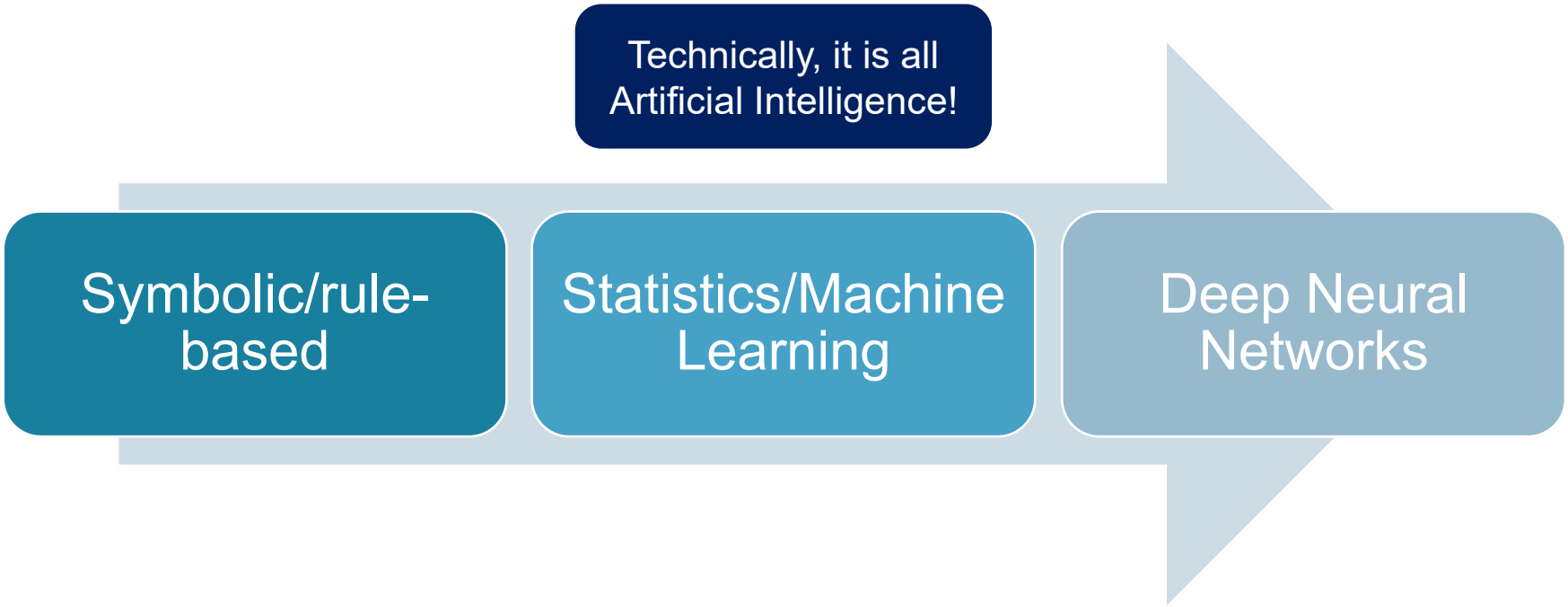
But manual annotation is painful!

1. Expensive

2. Time consuming

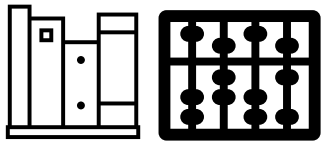
2. High quality of annotated data

Solution: automatic annotation

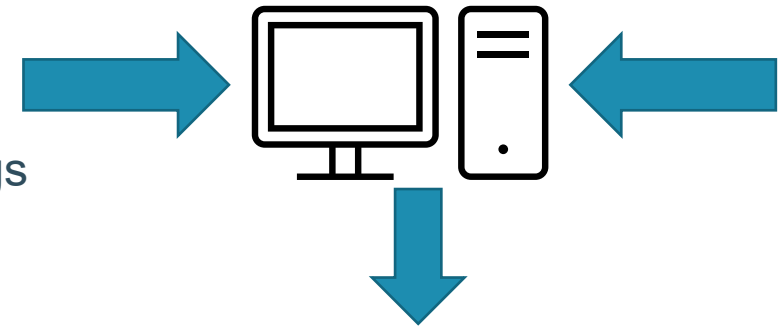


Symbolic or rule-based

Human experts design a **set of rules** that the machine applies to the data, and produces the analysis



- List of Latin words
- List of Latin word endings
- Inflectional rules
- Word-splitting rules



rosam

LEMMA: *rosa*
Morphology: First declension, accusative, singular, feminine

Examples for Latin: LEMLAT, Collatinus.

Note: Regular Expressions are an example of rule-based systems!

Statistical or Machine Learning (ML)

- The machine tries to identify **patterns/statistical rules** in the data
- Large variety of techniques, applying different statistical models and algorithms
- One very important distinction
  - **Supervised** learning
  - **Unsupervised** learning

Supervised vs Unsupervised

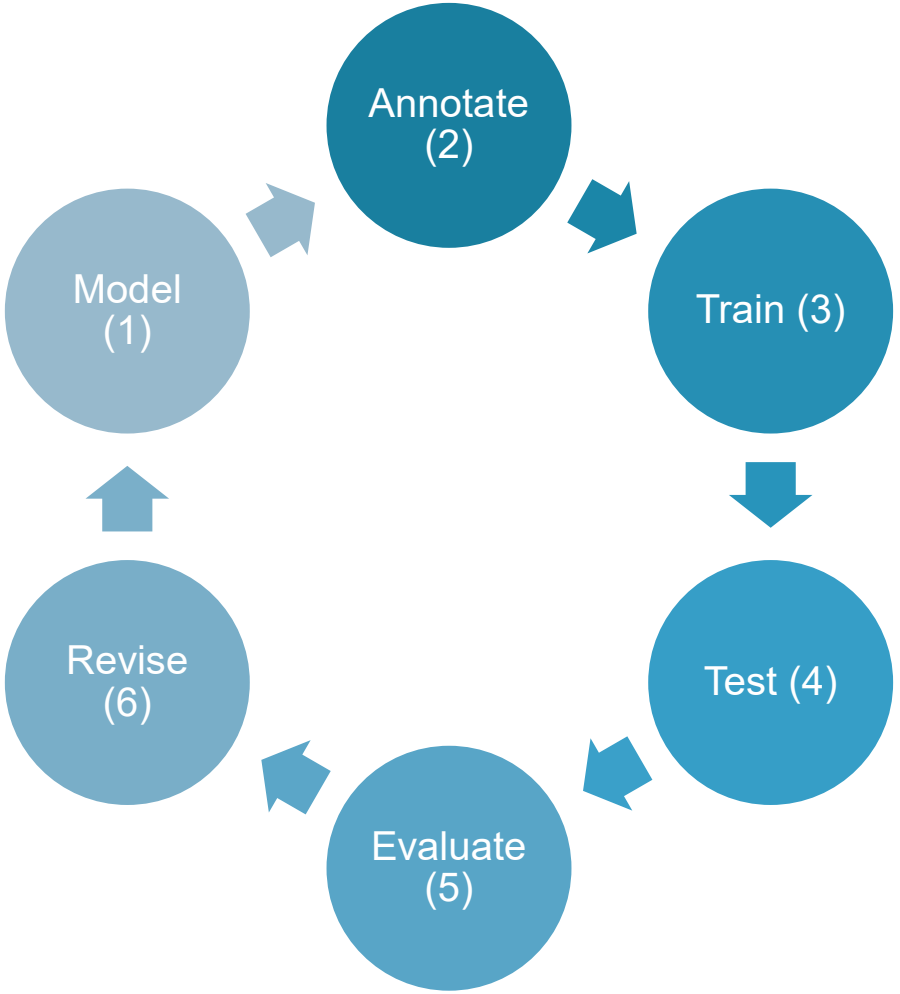
Supervised machine learning

The model is trained on **annotated** data to predict outputs on new text

Unsupervised machine learning

The model finds hidden patterns and structures in **unlabeled** data without predefined outputs

Supervised learning: a virtuous circle



Pustejovsky and Stubbs (2012) “Natural Language Annotation for Machine Learning”. O'Reilly Media.

(1) Model

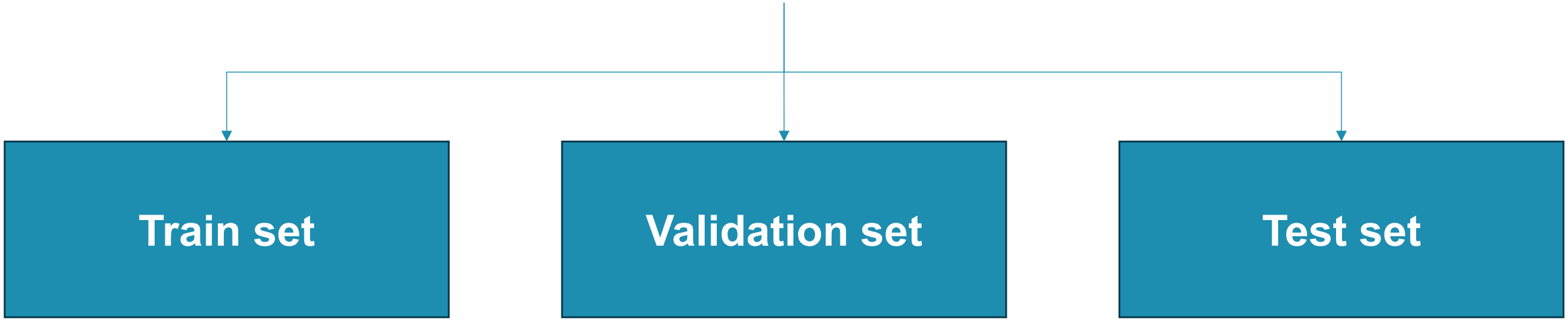
Decide what information you want to capture from your data and what statistical model to apply (e.g. morphological analysis)

- Example: Hidden Markov Models (HMM) → A text can be seen as a sequence of "**states**", which are in part **observable** (the words that we concretely see) and in part **unobservable** (their linguistic properties, for instance Part-of-Speech)
- HMM estimate, given a sequence of hidden labels, what is the likelihood that those specific words appear
- For instance the sequence ADJ – NOUN – VERB is more likely to generate *paruam pecuniam dat* than *quousque tandem Catilina.*

(2) Annotate

Annotate manually some of your data.

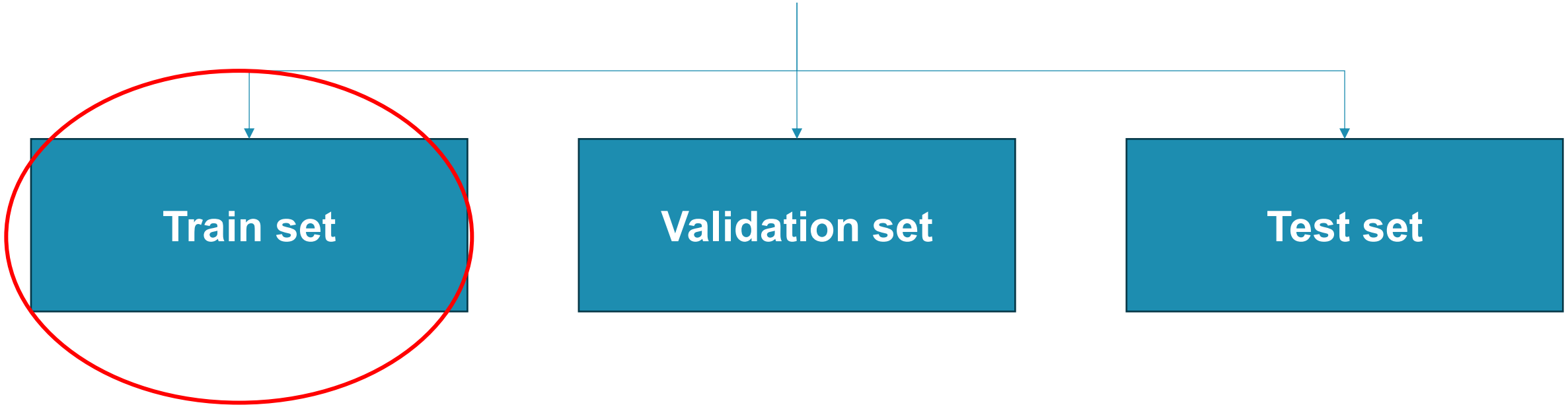
Very important: divide your corpus



(2) Annotate

Annotate manually some of your data.

Very important: divide your corpus



(2) Annotate

Annotate manually some of your data.

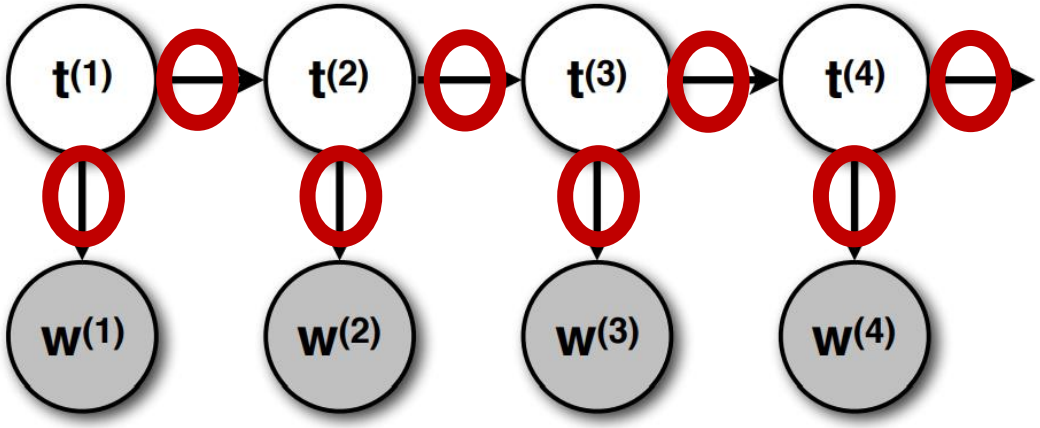
- Pick a formalism (for instance UD), annotation guidelines and **representative sample** of data to annotate (**train set**).
- Manually annotated data are called **GOLD data**: apart from human errors, they represent the optimal standard
- Ideally, annotated (gold) data are already available: for instance, the annotated corpora listed in the previous section
- However, inconsistencies in the annotated data are very dangerous → **harmonization**

(3) Train

Use the annotated data and the model so that the machine can infer patterns from your annotated data

By looking at the data, where each word is associated to its Part-of-Speech, **the program learns the probabilities** that link the Part-of-Speech and the words.

Computationally heavy

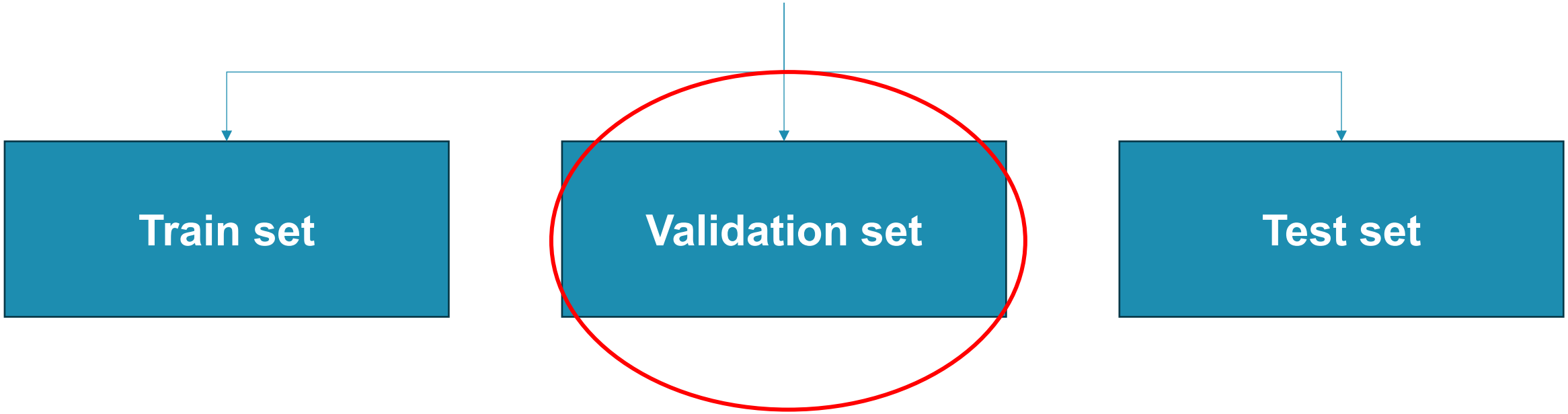


Often "**finetuning**" and not training from scratch

(3) Train

Use the annotated data and the model so that the machine can infer patterns from your annotated data

Very important: divide your corpus



(3) Train

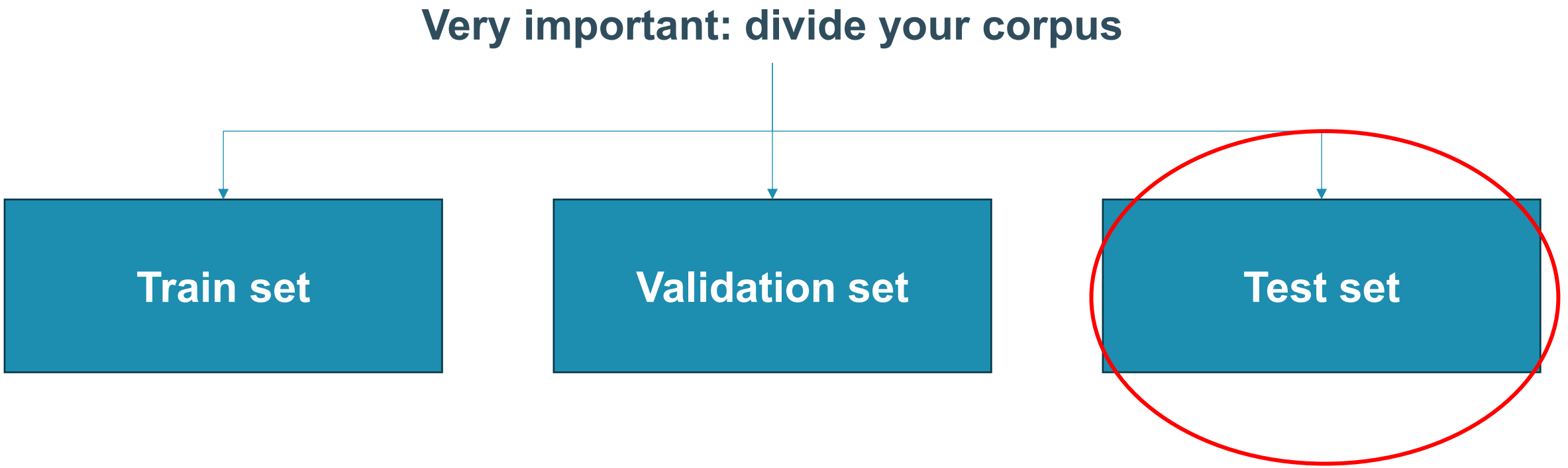
Use the annotated data and the model so that the machine can infer patterns from your annotated data

By looking at the data, where each word is associated to its Part-of-Speech, **the program learns the probabilities** that link the Part-of-Speech and the words.

- Risk of **overfitting**: the model becomes super efficient on the train set, but loses the ability to generalize on other texts.
- **Validation set**: used during the training process to check if the model is overfitting → if the performance on validation set gets worse, while the training performance keeps improving, you are overfitting the model.

(4) Test

Ask the machine to annotate unseen data, based on the patterns inferred.



(4) Test

Ask the machine to annotate unseen data, based on the patterns inferred.

- In this way, you assess whether the model is able to **generalize** beyond the data it was trained on.
- Use the trained model to annotate new, **unseen data**: not used during training (**test set**) and **not annotated**.
- This is needed to compare the annotation **PREDICTED** by your model and the **ACTUAL** annotation.

- E.g. GitHub - UniversalDependencies/UD_Latin-UDante

(5) Evaluate

Evaluate whether the performance is good.

- **Standardized** ways to evaluate the performance of the model
- Critical to provide it so that users can **pick the best model**
- Based on the comparison between the predicted values on the test set and the manually annotated values for the same text

Word	Predicted Part-of-Speech	Gold Part-of-Speech
tantae	ADJ	DET
ne	ADV	CONJ
animis	NOUN	NOUN

(5) Evaluate: confusion matrix

For each Part-of-Speech you count how the predicted value distributed

Example:

- You annotated 100 NOUNs
- For these words the model predicts 80 times NOUN, 10 ADJ, 5 DET and 5 VERB
- The confusion matrix captures this information

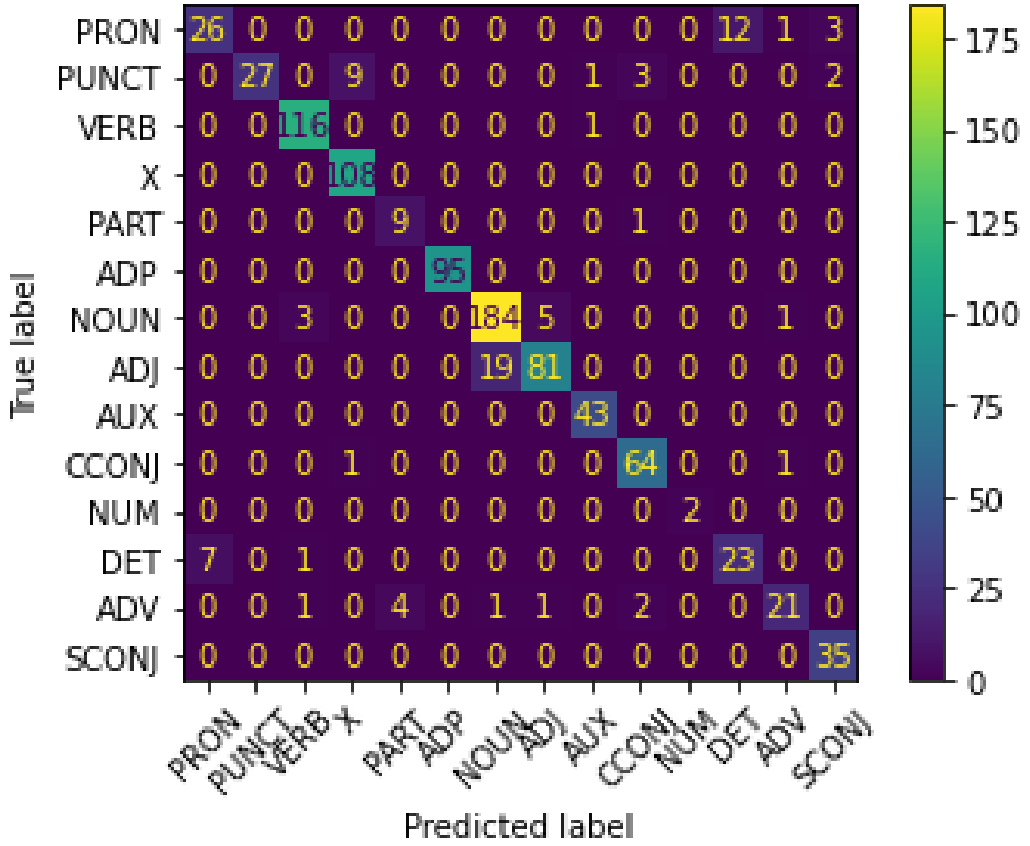
→ A way to visualize immediately the most problematic cases

predicted

gold

	NOUN	ADJ	PRON	DET	ADV	VERB	ADP
NOUN	80%	10%		5%		5%	
ADJ							
PRON							
DET							
ADV							
VERB							
ADP							

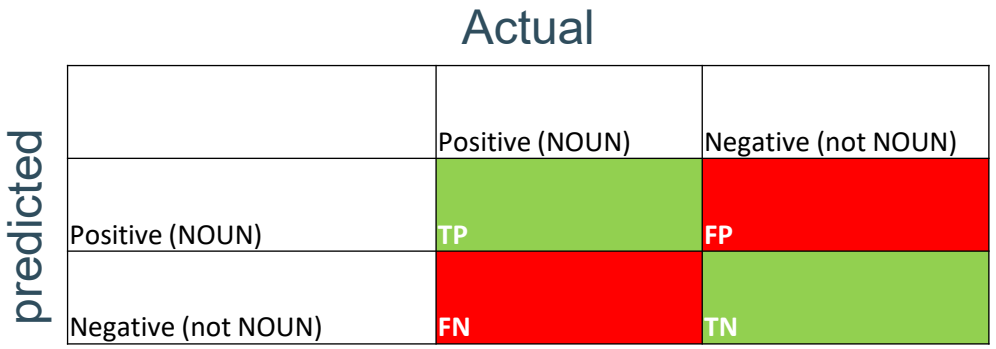
(5) Evaluate: confusion matrix



(5) Evaluate: per class evaluation metrics

For each class (here, Part-of-Speech) you can assess the quality of the prediction by looking at the true/false positive/negative

- E. g. if we want to evaluate the prediction of NOUNs we consider:
  - **TRUE POSITIVES (TP)**: how many NOUNS were predicted as NOUNS
  - **FALSE POSITIVES (FP)**: how many times NOUN was predicted when the word was not a NOUN
  - **TRUE NEGATIVES (TN)**: the cases in which non NOUNS were correctly predicted as non NOUNS
  - **FALSE NEGATIVES (FN)**: the cases in which NOUNs were labelled as something else



(5) Evaluation metrics: accuracy

ACCURACY:

ratio between the number of objects correctly analyzed and the total number of objects, i.e. how many times each category is predicted correctly over the total number of predictions.

$$\frac{TP + TN}{TP + TN + FP + FN}$$

- How many correct decisions were taken? How often is my model right?

Word	Predicted Part-of-Speech	Gold Part-of-Speech
tantae	ADJ	DET
Ne	ADV	CONJ
animis	NOUN	NOUN
caelestibus	NOUN	ADJ
irae	ADJ	NOUN
?	PUNCT	PUNCT

Example (NOUN category):

$$\frac{1\,TP + 3\,TN}{1\,TP + 3\,TN + 1\,FP + 1\,FN} = \frac{4}{6} = 0{,}66$$

(5) Evaluation metrics: accuracy

ratio b
analyz
how n
correct

Gold Part-of-Speech

DET
CONJ
NOUN
ADJ
NOUN
PUNCT

- How m
often is

):

$\frac{4}{6}$ = 0,66

DISCLAIMER

Accuracy alone can be misleading!

Example: the model correctly recognizes every other Part-of-Speech tags, but just a small part of the NOUNs. The general accuracy of the model will be very high, but high overall accuracy hides poor performance on specific Part-of-Speech tags

(5) Evaluation metrics: accuracy

ratio b
analyz
how n
correctly

We need more evaluation metrics

Gold Part-of-Speech
DET
CONJ
NOUN
ADJ
NOUN
PUNCT

- How m
often is

):

$\frac{4}{6}$ = 0,66

(5) Evaluation metrics: precision

PRECISION:

ratio between the number of correct predictions and the number of total predictions, i.e. how many times NOUN is predicted correctly over the total number of times in which it is predicted

$$\frac{TP}{TP + FP}$$

- Measures the **correctness** of the result
- Tip: the higher the number of **False Positives**, the lower the Precision.

Word	Predicted Part-of-Speech	Gold Part-of-Speech
tantae	ADJ	DET
ne	ADV	CONJ
animis	NOUN	NOUN
caelestibus	NOUN	ADJ
irae	ADJ	NOUN
?	PUNCT	PUNCT

Example : $\frac{1\ TP}{1\ TP + 1\ FP} = \frac{1}{2} = 0{,}5$

(5) Evaluation metrics: recall

RECALL:

ratio between the number of correct predictions and the effective number of nouns in the text, i.e. how many times NOUN is predicted correctly over the total number of NOUNs in the text

$$\frac{TP}{TP + FN}$$

- Measures the **completeness** of the result
- Tip: the higher the number of **False Negatives**, the lower the Recall.

Word	Predicted Part-of-Speech	Gold Part-of-Speech
Tantae	ADJ	DET
Ne	ADV	CONJ
Animis	NOUN	NOUN
Caelestibus	NOUN	ADJ
irae	ADJ	NOUN
?	PUNCT	PUNCT

Example : $\frac{1\ TP}{1\ TP + 1\ FN} = \frac{1}{2} = 0{,}5$

(5) Evaluation metrics: F1 score

F1 SCORE

combination of precision and recall → **harmonic mean** between precision and recall

$$\frac{2 \times Precision \times Recall}{(Precision + Recall)}$$

Example: $\frac{2 \times 0{,}5 \times 0{,}5}{0{,}5 + 0{,}5} = 0{,}5$

(6) Revise

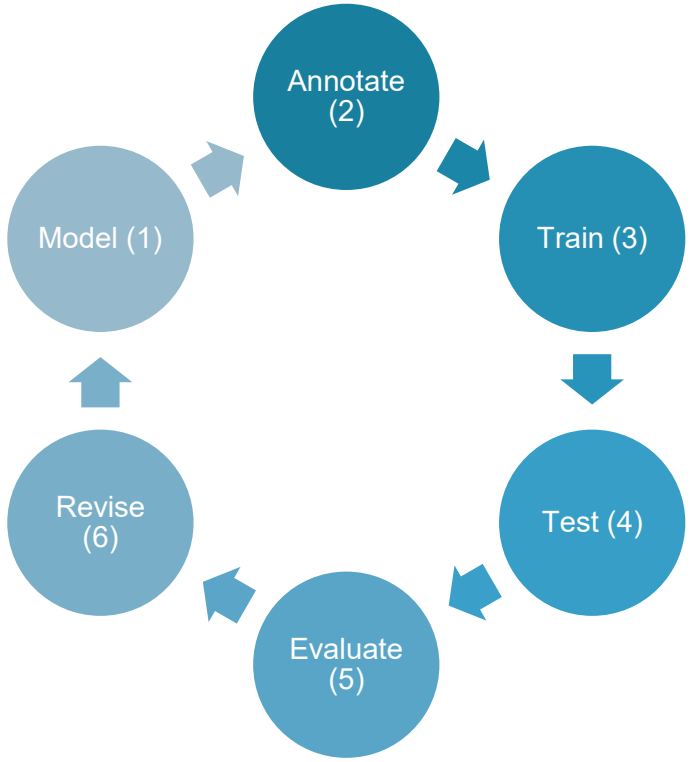
Revision of the model, of the annotation or of the algorithm.

Based on the evidence of the evaluation, you can apply changes to the model or to the annotation.

(6) Revise

Now you can quicky produce **more annotated data** → you can use them to train and enhance a new model!

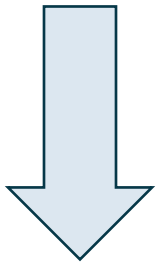
It's a **VIRTUOUS CIRCLE**.



Language models & Deep learning

Last step of automatic annotation evolution

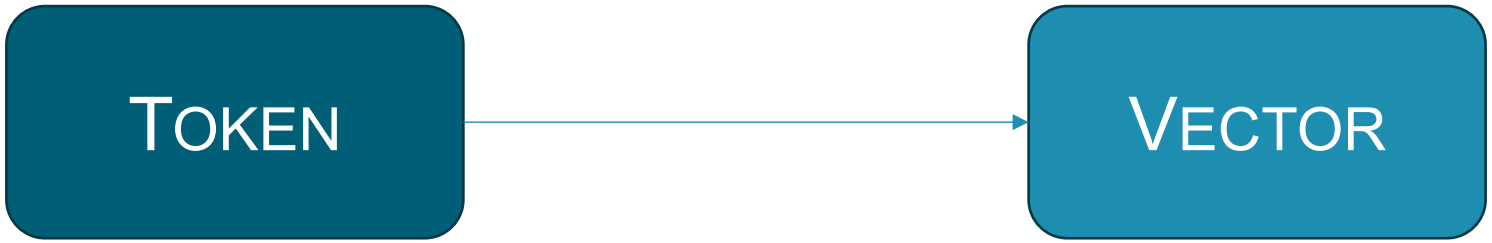
Architectures that can better capture long-distance relations in the sequence and bidirectional relations



TRANSFORMERS

Transformers

Transformers read all the input text at once and assign to each token a **vector** (**EMBEDDING**)

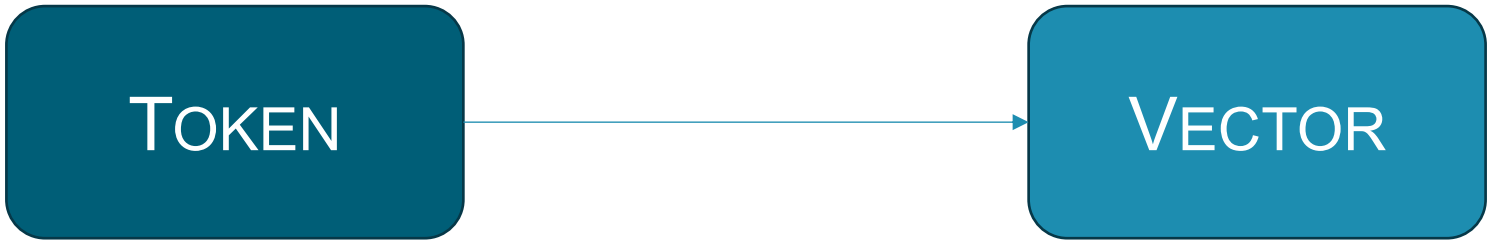


This vector is built based on the **contextual** information of the token.

Attention-mechanisms are used to capture the **most relevant** elements of the context

Transformers

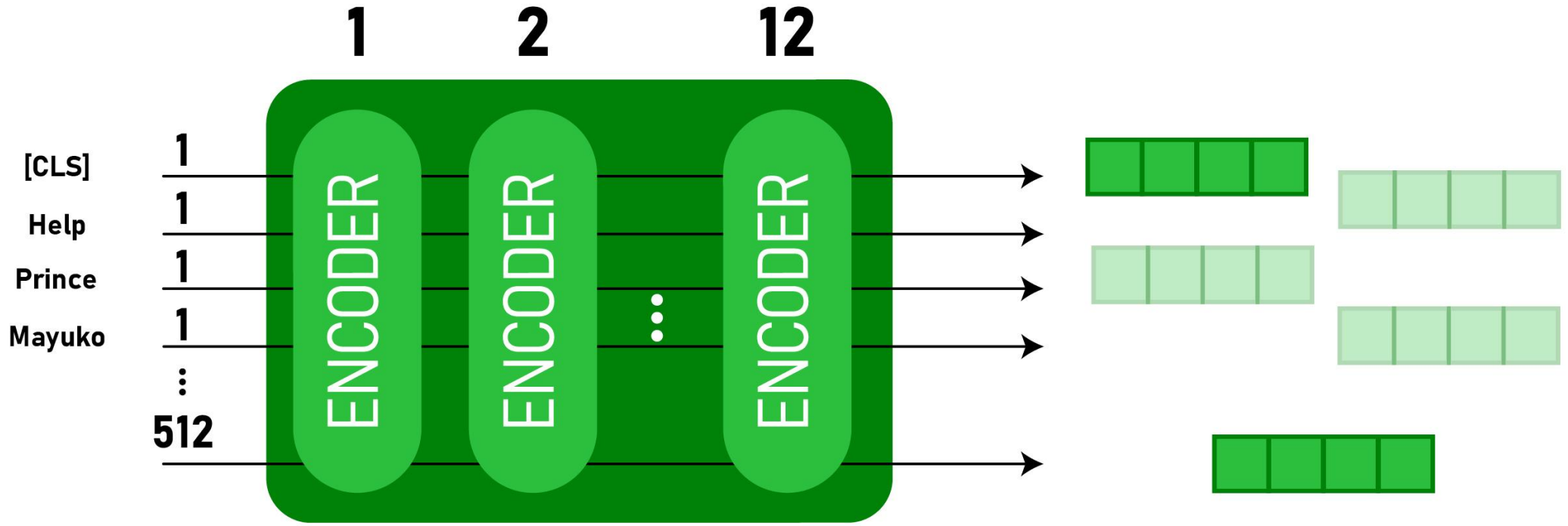
Transformers read all the input text at once and assign to each token a **vector** (**EMBEDDING**)



Essentially, tokens found in **similar contexts** are assigned **similar vectors**.

In this way, it is possible to capture the semantics of terms

Transformers



Explanation of BERT Model - NLP - GeeksforGeeks

What's important for us?

The described process is usually a task for the NLP developers. So… what can we do?

We can use the already available models and adapt them to our own project:

1. **Finetune** the pretrained model on your own data for your project-specific task.
2. Use the **pre-trained models**, ready to use.
   Note: It is very important to **always evaluate** the results!

   → Later in the practical session.

How can annotation be useful for my research?

Linguistically annotated text allows you to quickly find specific information on your text.

- Lemma search: finding all the times a word is used, without missing inflected or unpredictable forms (e.g. form *abeo* for the lemma *habeo* in LLCT).
  - ✓How does the vocabulary change across a text?
  - ✓How does the vocabulary change across different authors and time periods?
  - ✓Tracing lexical evolution and detecting thematic clusters

How can annotation be useful for my research?

Linguistically annotated text allows you to quickly find specific information on your text.

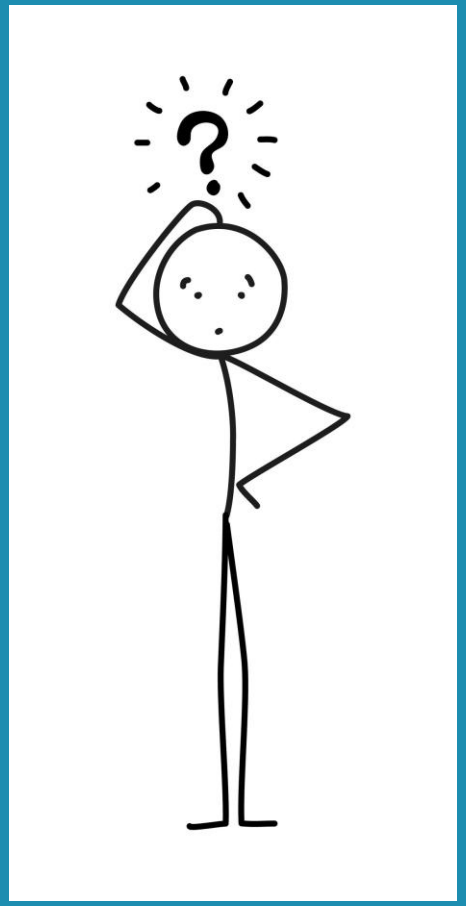
- Part-of-Speech and morphology: analyze how these are distributed and how they are related to the style of the authors
  - ✓Quantify stylistic markers (e.g. *nam* and other discourse markers)
  - ✓Compare Part-of-Speech patterns in different literary genres
  - ✓Compare how differ verbal tense and aspects (e.g. perfective vs imperfective) are used in different literary genres

How can annotation be useful for my research?

Linguistically annotated text allows you to quickly find specific information on your text.

- Syntactic annotation: compare structural differences between different texts
  - ✓Compare paratactic and hypotactic structures
  - ✓Analyze how word order changes across time periods (e.g. from SOV to SVO)
  - ✓Compare the structure of multi-lingual texts and translations (UD tags are standard for many languages!)

Any questions at this point?



How to annotate your own text?

Practical session

Try with your own corpus

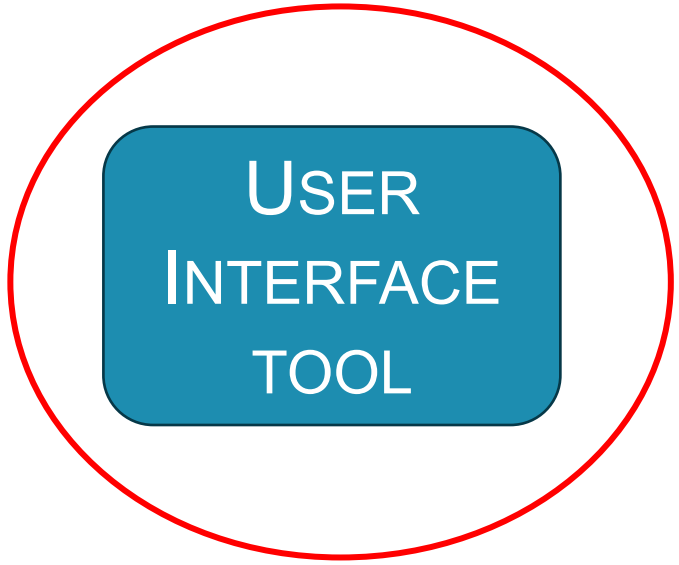
How to use a model?

USER INTERFACE TOOL

PYTHON NOTEBOOK

Try with your own corpus

How to use a model?



PYTHON NOTEBOOK

UD Pipe UI tool

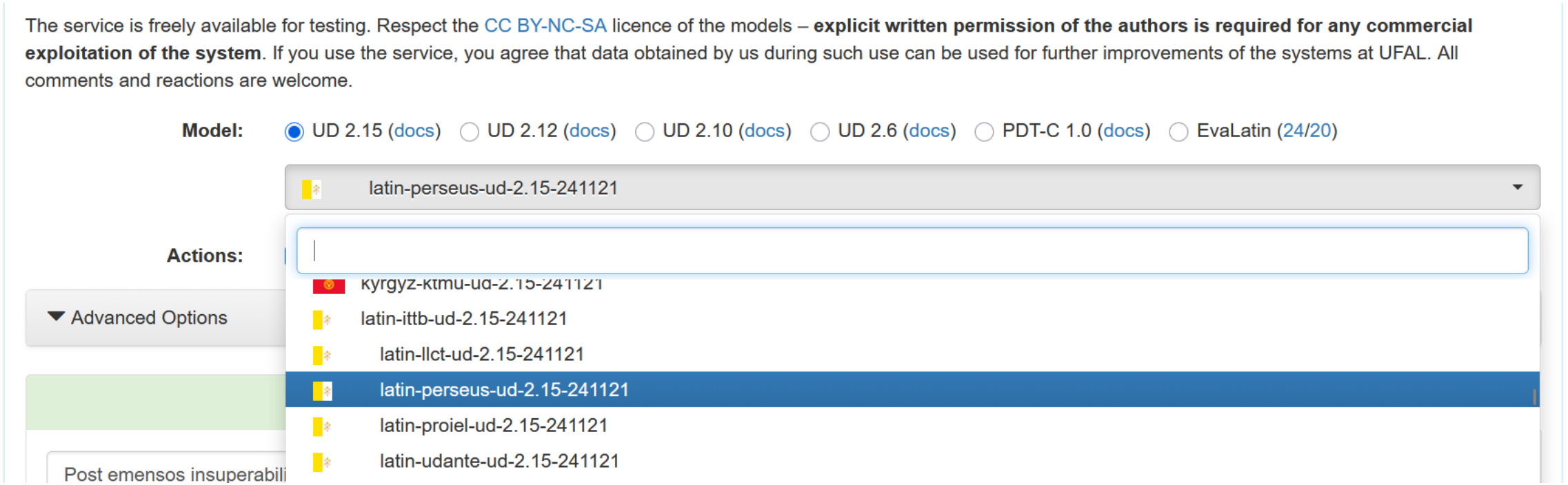
https://lindat.mff.cuni.cz/services/udpipe/

- You can paste your own text, or upload your own file
- You can choose what model to use
- You can choose the input format of the file
- You can download the results

UD Pipe UI tool

https://lindat.mff.cuni.cz/services/udpipe/

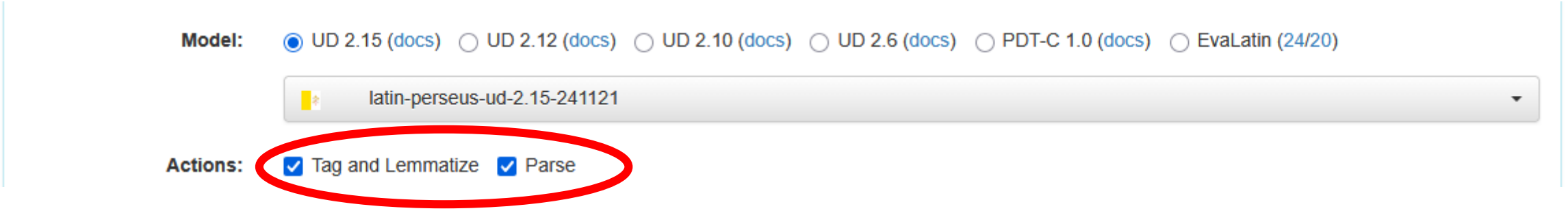
Choose what model to use



UD Pipe UI tool

https://lindat.mff.cuni.cz/services/udpipe/

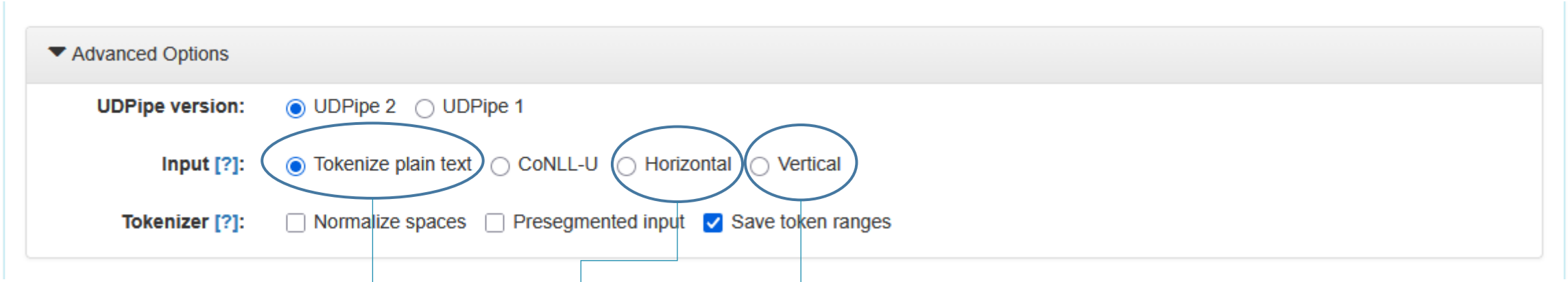
Choose what actions you want to perform



UD Pipe UI tool

https://lindat.mff.cuni.cz/services/udpipe/

Advanced options



Raw plain text

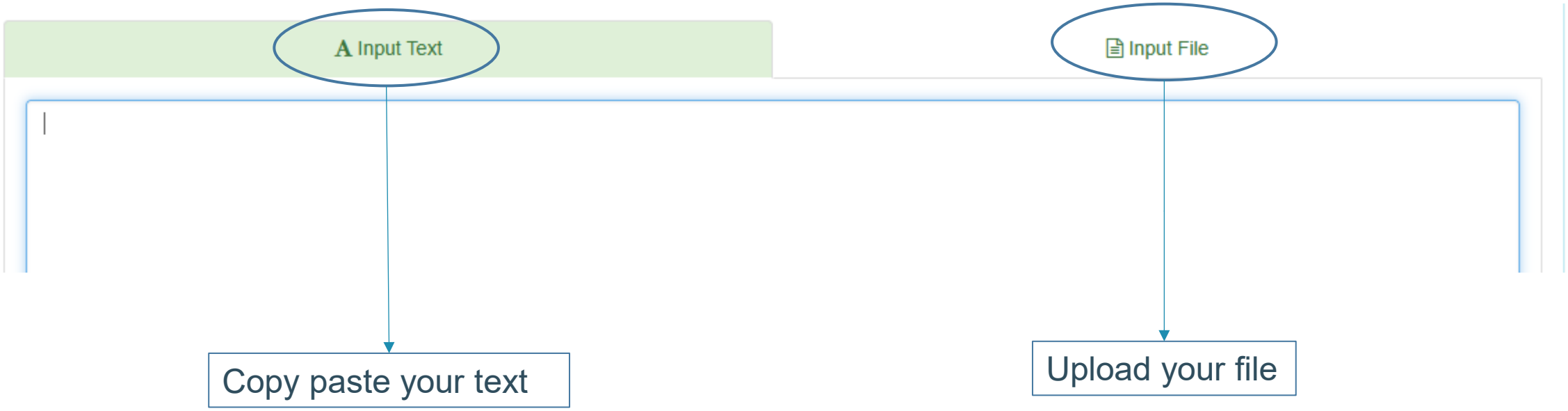
File is already split on sentences. Each sentence is on a different line

File is already tokenized. Each token is on a separate line, with an empty line denoting end of sentence

UD Pipe UI tool

https://lindat.mff.cuni.cz/services/udpipe/

Insert your text



Process Input

Output Text | Show Table | Show Trees

Save Output File

Id	Form	Lemma	UPosTag	XPosTag	Feats	Head	DepRel	Deps	Misc
# generator = UDPipe 2, https://lindat.mff.cuni.cz/services/udpipe									
# udpipe_model = latin-perseus-ud-2.15-241121									
# udpipe_model_licence = CC BY-NC-SA									
# newdoc									
# newpar									
# sent_id = 1									
# text = Gallia est omnis divisa in partes tres, quarum unam incolunt Belgae, aliam Aquitani, tertiam qui ipsorum lingua Celtae, nostra Galli appellantur.									
1	Gallia	Gallia	PROPN	n-s---fn-	Case=Nom|Gender=Fem|Number=Sing	4	nsubj:pass	_	TokenRange=0:6
2	est	sum	AUX	v3spia---	Aspect=Imp|Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin	4	aux:pass	_	TokenRange=7:10
3	omnis	omnis	DET	a-s---fn-	Case=Acc|Gender=Fem|Number=Plur|PronType=Tot	1	det	_	TokenRange=11:16
4	divisa	divido	VERB	v-srppfn-	Aspect=Perf|Case=Nom|Gender=Fem|Number=Sing|VerbForm=Part|Voice=Pass	0	root	_	TokenRange=17:23
5	in	in	ADP	r--------	_	6	case	_	TokenRange=24:26
6	partes	pars	NOUN	n-p---fa-	Case=Acc|Gender=Fem|Number=Plur	4	obl	_	TokenRange=27:33
7	tres	tres	NUM	C1|grn1|casM|gen2|vgr1	Case=Acc|Gender=Fem|Number=Plur|NumForm=Word|NumType=Card	6	nummod	_	SpaceAfter=No|TokenRange=34:38

You can visualize the resulting annotation as a CoNLLU file, or as a tree.

You can also save the output file

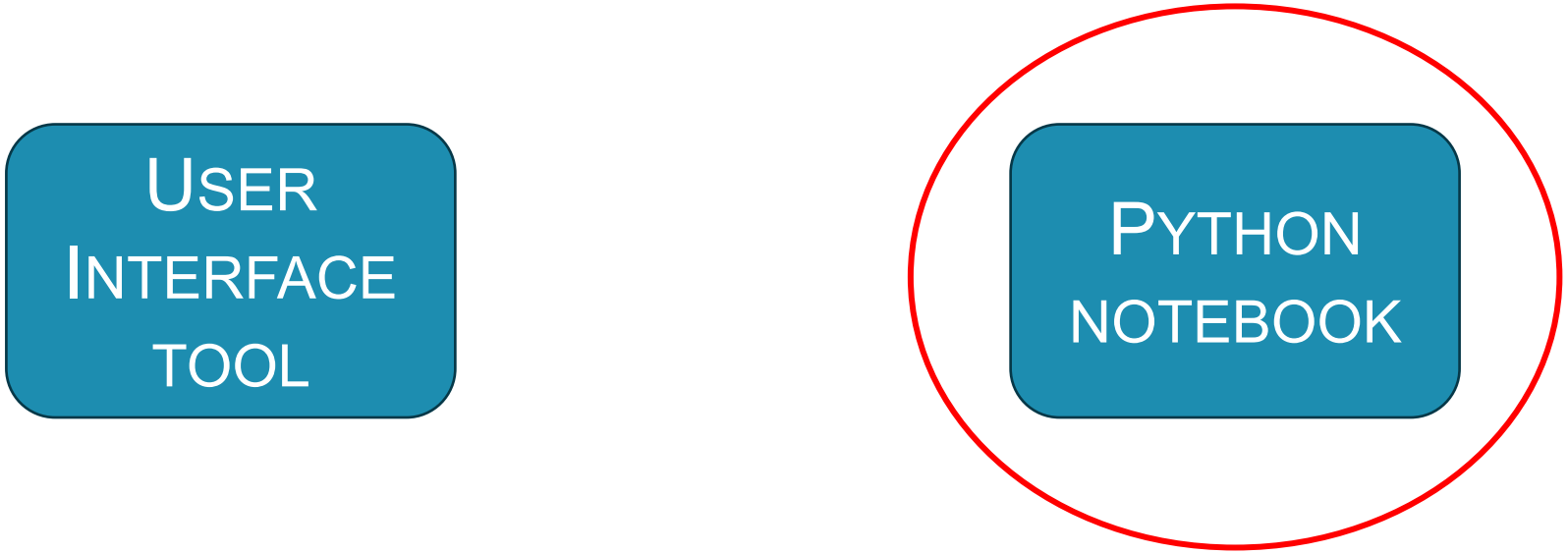
WebLicht chains

https://weblicht.sfs.uni-tuebingen.de/weblicht/

- Requires a subscription
- You can paste your own text, or upload your own file
- You can choose the models to combine your chain
- You can download the results

Try with your own corpus

How to use a model?



HuggingFace

https://huggingface.co/

- Platform for AI and machine learning
- Open-source community
- Offers pre-trained models: you can directly use them or finetune them for your task.
- Dataset library
- You can share your own model

Practical session: overview

- Perform the annotation on your own text:
  1. User interface: **UD Pipe tool**
  2. Python notebook: **Spacy**

  → Download the results in **CoNLLU** format

- Evaluation of the results:
  1. Focus on **Part-of-Speech tags**
  2. Add the correct annotation in an extra column
  3. Run the Python codebook for the evaluation

Practical session: details

- Perform the annotation on your own text:

1. You can use your own text you are working on.

2. You can use one of the standard text we provide:
    1. Book 1 of Caesar's *De Bello Gallico* (Latin)
    2. 20 sections of Plato's *Symposium* (Ancient Greek)
    3. 10 Chapters of Jane Austen's *Pride and Prejudice* (English)
    4. 3 Chapters of Alexandre Dumas' *Le comte de Monte Cristo* (French)

Practical session: details

- Perform the annotation on your own text:

    1. UD Pipe UI tool: choose a model that works on your selected language and follow the instructions

    2. Python notebook:
        1. LatinCy.ipynb → works on Latin
        2. GreCy.ipynb → works on Ancient Greek
        3. SpaCY_english_French.ipynb → works on English and French

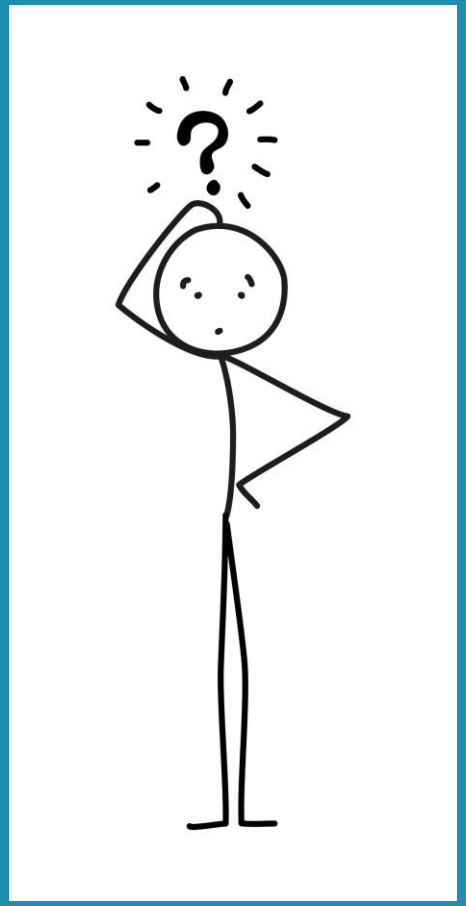
Practical session: details

Evaluation of the results

Follow the detailed instructions available on the Google Drive shared folder. This is required for making the notebook work correctly.

1. Focus on **Part-of-Speech tags**
2. Open the file in Excel
3. Add the correct Part-of-Speech tag in an extra column (“Gold_UPOS”)
4. Save the file
5. Run the Python codebook for the evaluation
6. Discuss the results

Any questions at this point?



Good luck!

